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□□□□□□□□□□□A2G/U2U□□□□□□□□□LoS□□□□□□□□□□□□□□□□□□□

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1.1 □□□□□□□□□□

□□□□□□□□ ScenarioGenerator □□□□□□□[1]□□□□□□□□□□□□□□□□□□□□□□□□ area_size □□□□□□□□□□□□□□□□□road/sidewalk/roof/indoor □□□□□□□□□□□□□□□□□□□□□□□□□□□□□area_size□ building_density□building_height_range□environment_type □□

1.2 □□ α–β–γ □□□□□□□□□

□□□□ Al-Hourani □□□□□□□□□□□□α, β, γ□□□□□□□□□dense_urban/urban/suburban/rural□□□□ α □□□□□□□□β □□□□□□□buildings/km²□□γ □□□□□ Rayleigh □□□□□□□□□□□□□□□□□□□□□□ AL_HOURANI_ENV_PARAMS □□□□[2][1]

1.3 □□□□ W □□□□□ S □□□

□□ α □ β□□□□□□□□□□□□□□□□□□□ W □□□□□□□□ S□

$$W = 1000 * \mathrm{sqrt}(\alpha / \beta) \quad (1)$$

$$S = 1000 / \mathrm{sqrt}(\beta) - W \quad (2)$$

□□□□□ _generate_building_params □□□□□□□□□□□ footprint □□□□□/□□□□□□□□□□[1][2]

1.4 □□□□□□□□□□□□

□□□□□□□□□□ β □□□

$$N_build = \mathrm{floor}(area_size^2 * \beta / 10^6) \quad (3)$$

□□□□□□□□□□□□□□□□□□□

```
1028:
1029:             # □□□□
```

□□□ A2G/U2U/□□□-□□□□□□□□□□□□□

2.1 □□□□□□□□□FSPL□

□□□□□□□□□□□□□□□□□□□calculate_fspl_db□□□□□ Friis □□□□[1]

$$FSPL(d,f)=20\cdot\log10(4\pi\ d\ /\ \lambda)\quad(4)$$

2.2 A2G □□□□□□□LoS/NLoS □□□□

□ A2G □□□□□□□ Al-Hourani □□□[2][1]

$$L_A2G = FSPL + P_LoS\cdot\eta_LoS + (1-P_LoS)\cdot\eta_NLoS\quad(5)$$

2.3 LoS □□□□□□□□□□ Sigmoid □□

$$P_LoS = \Pi_{i=1}^{m}\ (1 - \exp(-z_i^2/(2\gamma^2)))\quad(6)$$

$$P_LoS(\theta) = 1/(1 + a\cdot\exp(-b(\theta-a)))\quad(7)$$

□□□□□□□ Al-Hourani □□□□□□□□□□□ calculate_los_probability □□□□[2][1]

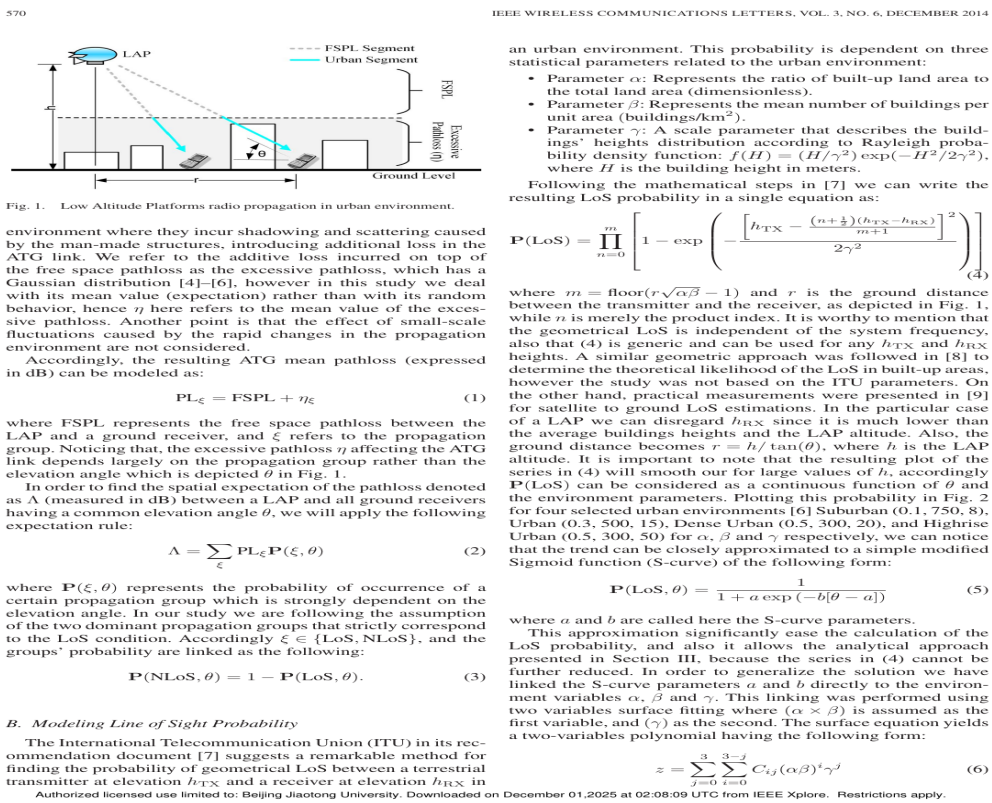

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Fig. 1. Low Altitude Platforms radio propagation in urban environment.

environment where they incur shadowing and scattering caused by the man-made structures, introducing additional loss in the ATG link. We refer to the additive loss incurred on top of the free space pathloss as the excessive pathloss, which has a Gaussian distribution [4]–[6], however in this study we deal with its mean value (expectation) rather than with its random behavior, hence η here refers to the mean value of the excessive pathloss. Another point is that the effect of small-scale fluctuations caused by the rapid changes in the propagation environment are not considered.

Accordingly, the resulting ATG mean pathloss (expressed in dB) can be modeled as:

$$\mathrm{PL}_{\xi} = \mathrm{FSPL} + \eta_{\xi} \quad (1)$$

where FSPL represents the free space pathloss between the LAP and a ground receiver, and ξ refers to the propagation group. Noticing that, the excessive pathloss η affecting the ATG link depends largely on the propagation group rather than the elevation angle which is depicted θ in Fig. 1.

In order to find the spatial expectation of the pathloss denoted as Λ (measured in dB) between a LAP and all ground receivers having a common elevation angle θ, we will apply the following expectation rule:

$$\Lambda = \sum_{\xi} \mathrm{PL}_{\xi}\mathbf{P}(\xi,\theta) \quad (2)$$

where $\mathbf{P}(\xi,\theta)$ represents the probability of occurrence of a certain propagation group which is strongly dependent on the elevation angle. In our study we are following the assumption of the two dominant propagation groups that strictly correspond to the LoS condition. Accordingly $\xi \in \{\mathrm{LoS}, \mathrm{NLoS}\}$, and the groups' probability are linked as the following:

$$\mathbf{P}(\mathrm{NLoS},\theta) = 1 - \mathbf{P}(\mathrm{LoS},\theta). \quad (3)$$

B. Modeling Line of Sight Probability

The International Telecommunication Union (ITU) in its recommendation document [7] suggests a remarkable method for finding the probability of geometrical LoS between a terrestrial transmitter at elevation h_{TX} and a receiver at elevation h_{RX} in an urban environment. This probability is dependent on three statistical parameters related to the urban environment:

- Parameter α: Represents the ratio of built-up land area to the total land area (dimensionless).
- Parameter β: Represents the mean number of buildings per unit area (buildings/km^2).
- Parameter γ: A scale parameter that describes the buildings' heights distribution according to Rayleigh probability density function: $f(H) = (H/\gamma^2)\exp(-H^2/2\gamma^2)$, where H is the building height in meters.

Following the mathematical steps in [7] we can write the resulting LoS probability in a single equation as:

$$\mathbf{P}(\mathrm{LoS}) = \prod_{n=0}^{m}\left[1 - \exp\left(-\frac{\left[h_{\mathrm{TX}} - \frac{(n+\frac{1}{2})(h_{\mathrm{TX}}-h_{\mathrm{RX}})}{m+1}\right]^2}{2\gamma^2}\right)\right] \quad (4)$$

where $m = \mathrm{floor}(r\sqrt{\alpha\beta} - 1)$ and r is the ground distance between the transmitter and the receiver, as depicted in Fig. 1, while n is merely the product index. It is worthy to mention that the geometrical LoS is independent of the system frequency, also that (4) is generic and can be used for any h_{TX} and h_{RX} heights. A similar geometric approach was followed in [8] to determine the theoretical likelihood of the LoS in built-up areas, however the study was not based on the ITU parameters. On the other hand, practical measurements were presented in [9] for satellite to ground LoS estimations. In the particular case of a LAP we can disregard h_{RX} since it is much lower than the average buildings heights and the LAP altitude. Also, the ground distance becomes $r = h/\tan(\theta)$, where h is the LAP altitude. It is important to note that the resulting plot of the series in (4) will smooth our for large values of h, accordingly $\mathbf{P}(\mathrm{LoS})$ can be considered as a continuous function of θ and the environment parameters. Plotting this probability in Fig. 2 for four selected urban environments [6] Suburban (0.1, 750, 8), Urban (0.3, 500, 15), Dense Urban (0.5, 300, 20), and Highrise Urban (0.5, 300, 50) for α, β and γ respectively, we can notice that the trend can be closely approximated to a simple modified Sigmoid function (S-curve) of the following form:

$$\mathbf{P}(\mathrm{LoS},\theta) = \frac{1}{1 + a\exp\left(-b[\theta - a]\right)} \quad (5)$$

where a and b are called here the S-curve parameters.

This approximation significantly ease the calculation of the LoS probability, and also it allows the analytical approach presented in Section III, because the series in (4) cannot be further reduced. In order to generalize the solution we have linked the S-curve parameters a and b directly to the environment variables α, β and γ. This linking was performed using two variables surface fitting where $(\alpha \times \beta)$ is assumed as the first variable, and (γ) as the second. The surface equation yields a two-variables polynomial having the following form:

$$z = \sum_{j=0}^{3}\sum_{i=0}^{3-j} C_{ij}(\alpha\beta)^i\gamma^j \quad (6)$$

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□□LoS □□□ A2G □□□□ [2]

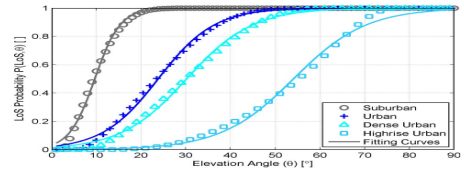

Fig. 2. The calculated line-of-sight probabilities, with their related S-curve fitting for different urban environments.

TABLE I
SURFACE POLYNOMIAL COEFFICIENTS FOR a

C_{ij}	i	0	1	2	3
j					
0		9.34E-01	2.30E-01	-2.25E-03	1.86E-05
1		1.97E-02	2.44E-03	6.58E-06	-
2		-1.24E-04	-3.34E-06	-	-
3		2.73E-07	-	-	-

TABLE II
SURFACE POLYNOMIAL COEFFICIENTS FOR b

C_{ij}	i	0	1	2	3
j					
0		1.17E+00	-7.56E-02	1.98E-03	-1.78E-05
1		-5.79E-03	1.81E-04	-1.65E-06	-
2		1.73E-05	-2.02E-07	-	-
3		-2.00E-08	-	-	-

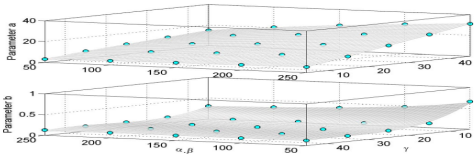

Fig. 3. S-curve parameters 3D-fitting as a relation to the urban environment parameters.

where z represents the fitting parameter a or b, and C_{ij} are the polynomial coefficients given in Table I and Table II, while the surface fitting is depicted in Fig. 3.

III. FINDING THE OPTIMUM ALTITUDE

In order to analyze the effect of the LAP's altitude on the provided service, firstly we define the service threshold in terms of the maximum allowable pathloss $\mathrm{PL}_{\max}$. When the total pathloss between the LAP and a receiver exceeds this threshold,

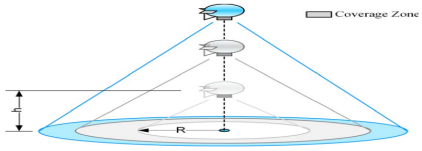

Fig. 4. The coverage zone by a low altitude platform.

the link is deemed as failed. For ground receivers, this threshold translates into a coverage disk (zone) of radius R, since all receivers within this disk have a pathloss that is less than or equal $\mathrm{PL}_{\max}$, as depicted in Fig. 4. Mathematically speaking, the cell radius of the coverage zone can be written as:

$$R = r|_{\Lambda=\mathrm{PL}_{\max}} \tag{7}$$

Accordingly, the optimization problem is to find the best altitude that will maximize R. In order to do so, we deduce a relation between the LAP altitude h and the cell radius R. By rewriting Equation (1) we have:

$$\begin{aligned}\mathrm{PL}_{\mathrm{LoS}} &= 20\log d + 20\log f + 20\log\left(\frac{4\pi}{c}\right) + \eta_{\mathrm{LoS}}\\ \mathrm{PL}_{\mathrm{NLoS}} &= \underbrace{20\log d + 20\log f + 20\log\left(\frac{4\pi}{c}\right)}_{\mathrm{FSPL}} + \underbrace{\eta_{\mathrm{NLoS}}}_{\eta_\epsilon}\end{aligned} \tag{8}$$

where d is the distance between the LAP and a receiver at a circle of radius r, given by $d = \sqrt{h^2 + r^2}$, while f is the system frequency. The FSPL is according to Friis equation with the assumption of isotropic transmitter and receiver antennas. Referring to (2):

$$\Lambda = \mathbf{P}(\mathrm{LoS}) \times \mathrm{PL}_{\mathrm{LoS}} + \mathbf{P}(\mathrm{NLoS}) \times \mathrm{PL}_{\mathrm{NLoS}} \tag{9}$$

According to Fig. 1, we notice that $\theta = \arctan(h/r)$. Now we substitute from (3), (5), (7), (8) into (9), then performing some simple algebraic reductions, we can write:

$$\mathrm{PL}_{\max} = \frac{A}{1 + a\exp\left(-b\left[\arctan\left(\frac{h}{R}\right) - a\right]\right)} + 10\log(h^2 + R^2) + B \tag{10}$$

where $A = \eta_{\mathrm{LoS}} - \eta_{\mathrm{NLoS}}$ and $B = 20\log f + 20\log(4\pi/c) + \eta_{\mathrm{NLoS}}$. The above equation is implicit, where neither R nor h can be written as an explicit function of each other. In order to obtain the optimum point of the LAP altitude h_{OPT} that yields the best coverage, we need to search for the value of h that satisfies the equation of the critical point:

$$\frac{\partial R}{\partial h} = 0 \tag{11}$$

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□□A2G □□□□□□□ [2]

2.5 □□□□□□□□□□□-□□□□AABB□□□

□□□□ AABB□□□□□□□ slabs method□□ x/y/z □□□□□□□□ t □□□□□□□□□□□□□□□□□□□ Williams □□[8][1]

```
    def intersects_segment_3d(self, p0: np.ndarray, p1: np.ndarray, eps: float = 1e-6) -> bool:
        """
        □□ 3D □□ p0->p1 □□□□□□□□□□□□□□□□□□□□□□□□ z=0□□□□ self.height□□

        □□□
        - □□□□□□□□□□□□□□□□□□□□ AABB□
        - □□□"□□□□□□□□□□"□□□□□□□□ eps □□ t=0/1 □□□□□□□□
        """
        p0 = np.asarray(p0, dtype=float)
        p1 = np.asarray(p1, dtype=float)
        d = p1 - p0

        xmin = float(self.x - self.width / 2)
        xmax = float(self.x + self.width / 2)
        ymin = float(self.y - self.length / 2)
        ymax = float(self.y + self.length / 2)
        zmin = 0.0
        zmax = float(self.height)

        tmin = 0.0
        tmax = 1.0

        for axis, (bmin, bmax) in enumerate(((xmin, xmax), (ymin, ymax), (zmin, zmax))):
            if abs(float(d[axis])) < 1e-12:
                # □□□□□□□□□□□□ slab □□□□□
                if float(p0[axis]) < bmin or float(p0[axis]) > bmax:
                    return False
                continue

            inv = 1.0 / float(d[axis])
            t1 = (bmin - float(p0[axis])) * inv
            t2 = (bmax - float(p0[axis])) * inv
            t_enter = min(t1, t2)
            t_exit = max(t1, t2)
```

```
            tmin = max(tmin, t_enter)
            tmax = min(tmax, t_exit)
            if tmax < tmin:
                return False

        # □□□□□□□□□□□□□□□□□ (eps, 1-eps) □
        return (tmax > eps) and (tmin < 1.0 - eps)

    def get_suitable_interference_heights(self, interference_type: str) -> List[float]:
        """□□□□□□□□□□□□□□□□□"""
        height_distribution = {
            'ground': [2],  # □□
```

2.6 □□□P.1238□□□□□□□□□□P.2109□

□□□□□□□□□□□□ ITU-R P.1238 □□□□□ ITU-R P.2109 □□□□□□□□[3][4][1]

3 Basic transmission loss models

The use of this indoor transmission loss model assumes that the base station and portable terminal are located inside the same building. The indoor base to mobile/portable radio basic transmission loss can be estimated with either site-general or site-specific models.

3.1 Site-general models

The site-general model is applicable to situations where both the transmitting and receiving stations are located on the same floor. The median basic transmission loss is given by:

$$L_b(d, f) = 10\alpha \log_{10}(d) + \beta + 10\gamma \log_{10}(f) \quad \text{dB} \qquad (1)$$

with an additive zero mean Gaussian random variable $N(0, \sigma)$ with a standard deviation σ (dB), where:

- d : 3D direct distance between the transmitting and receiving stations (m)
- f : operating frequency (GHz)
- α : coefficient associated with the increase of the basic transmission loss with distance
- β : coefficient associated with the offset value of the basic transmission loss
- γ : coefficient associated with the increase of the basic transmission loss with frequency.

For non-line-of-sight (NLoS) Monte Carlo simulations, the excess basic transmission loss with respect to free-space basic transmission loss, L_{FS} ($L_{FS} = 20 \log_{10}(4 \times 10^9 \pi d f / c)$ where c is the speed of light in m/s), will not exceed $10 \log_{10}(10^{0.1A} + 1)$ (dB), where A is a random variable with a normal distribution $N(\mu, \sigma)$, with mean $\mu = L_b(d, f) - L_{FS}$ and standard deviation of σ.

The recommended coefficient values for indoor propagation environments are provided in Table 2.

TABLE 2

Basic transmission loss coefficients

Environment	LoS/NLoS	Frequency range (GHz)	Distance range (m)	α	β	γ	σ
Office	LoS	0.3-294.0	2-27	1.47	34.17	2.08	3.68
	NLoS	0.3-255.0	4-30	2.39	30.13	2.40	5.01
Corridor	LoS	0.3-300.0	2-160	1.57	29.46	2.24	3.77
	NLoS	0.625-159.0	3-94	2.78	28.62	2.54	7.58
Industrial	LoS	0.625-294.0	2-102	2.27	24.79	2.10	2.62
	NLoS	0.625-255.0	3-110	2.80	23.55	2.16	5.70
Conference/lecture room	LoS	0.45-300.0	2-21	1.56	30.47	2.23	2.92
	NLoS	0.45-159.0	4-25	1.40	39.53	2.37	3.33

3.2 Site-specific models

For estimating the basic transmission loss or field strength, site-specific models are also useful. Models for indoor field strength prediction based on the uniform theory of diffraction (UTD) and ray-tracing techniques are available. Detailed information of the building structure is necessary for the calculation of the indoor field strength. These models combine empirical elements with the

□□ITU-R P.1238 □□□□□□□□□□ [3]

The model makes the implicit assumption that terminals have an equal probability of location at any point within a building.

2.1 Classification of building type

Experimental results, such as those collated in Report ITU-R P.2346, shows that, when characterised in terms of entry loss, buildings fall into two distinct populations: where modern, thermally-efficient building methods are used (metallised glass, foil-backed panels) building entry loss is generally significantly higher than for 'traditional' buildings without such materials. The model therefore gives predictions for these two cases.

This classification, of 'thermally-efficient' and 'traditional', refers purely to the thermal efficiency of construction materials. No assumption should be made on the year of construction, type (single or multi-floors), heritage or building method.

For building entry loss, it is important to consider the thermal efficiency of the complete building (or the overall thermal efficiency). A highly thermally-efficient main structure with poorly insulated windows (e.g. single glazed with thin glass) can make the building thermally-inefficient and vice versa.

Thermal transmittance, commonly referred as U-value, provides a quantifiable description of thermal efficiency. Low U-values represent high thermal efficiency. Typically, the presence of metallised glass windows, insulated cavity walls, thick reinforced concrete and metal foil back cladding is a good indication[1] of a thermally-efficient building.

3 Model

Building entry loss will vary depending on building type, location within the building and movement in the building. The building entry loss distribution is given by a combination of two lognormal distributions. The building entry loss not exceeded for the probability, P, is given by:

$$L_{BEL}^{omni}(P) = 10\log(10^{0.1A(P)} + 10^{0.1B(P)} + 10^{0.1C}) \text{ dB} \tag{1}$$

where:

$$A(P) = F^{-1}(P)\sigma_1 + \mu_1 \tag{2}$$

$$B(P) = F^{-1}(P)\sigma_2 + \mu_2 \tag{3}$$

$$C = -3.0 \tag{4}$$

$$\mu_1 = L_h + L_e \tag{5}$$

$$\mu_2 = w + x\log(f) \tag{6}$$

$$\sigma_1 = u + v\log(f) \tag{7}$$

$$\sigma_2 = y + z\log(f) \tag{8}$$

where:

L_h : median loss for horizontal paths, given by:

$$L_h = r + s\log(f) + t(\log(f))^2 \tag{9}$$

L_e : correction for elevation angle of the path at the building façade:

---

[1] For example, U-values of < 0.3 and < 0.9 are representative of thermally efficient main structure and metallised glass, respectively.

□□ITU-R P.2109 □□□□□□□□ [4]

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3.1 □□□□□□□

□□□□ InterferenceSourceConfig/InterferenceSource □□□□□□□□□□□□□□□□□□□□

1) 你城市生成模型的可观测量

对每栋楼 i（你代码里有 W_i, L_i, H_i）：

- 体积

$$V_i = W_i L_i H_i$$

- footprint

$$A_i^{\text{foot}} = W_i L_i$$

- 层数（用你代码已有 `floor_height_m`）

$$n_i = \max\left(1, \left\lfloor \frac{H_i}{h_{\text{floor}}} \right\rfloor\right)$$

- 总楼面面积（这一步与 wardriving 文中"footprint×floors"一致）

$$A_i^{\text{floor}} = A_i^{\text{foot}}\, n_i$$

> 这一步是关键：虽然你起点是"体积"，但最终推"户/办公室数量"最自然的中间量是**楼面面积**（否则"每户体积"还得除以层高才能等价成面积）。论文在 business 侧就是把**楼面面积**作为承载量来分配并推 AP 数的。 Wi-Fi Wardriving Studies Must A...

□□□□□□.docx □□ [17]

2) 最简单的"体积驱动 Wi-Fi 密度"模型（Residential + Business）

把 Wi-Fi AP 数分两部分（论文也明确这么拆）： Wi-Fi Wardriving Studies Must A...

$$N_{\text{wifi}} = N_{\text{res}} + N_{\text{bus}}$$

2.1 Residential（每户一路由器/每户 Wi-Fi 采用率）

假设住宅楼面中，平均每户面积为 A_{dwell}（m²/户），住宅楼面占比为 $\eta_{\text{res}} \in [0,1]$，每户平均路由器数为 κ_{res}（通常≈1，可作为参数），Wi-Fi 采用率为 p_{wifi}（论文里用"宽带采用→Wi-Fi 采用"的概率链来估计，这里简化成一个合并概率）。 Wi-Fi Wardriving Studies Must A...

先由楼面估计户数：

$$N_{\text{dwell}} = \sum_i \frac{\eta_{\text{res}} A_i^{\text{floor}}}{A_{\text{dwell}}}$$

住宅 Wi-Fi AP 数：

$$N_{\text{res}} = \kappa_{\text{res}}\, p_{\text{wifi}}\, N_{\text{dwell}}$$

> 这一步就是你想要的"每户体积/每户数量→Wi-Fi 数"。
> 若你坚持"每户体积 V_{dwell}"而不是面积，也可以等价写成：
> $A_{\text{dwell}} = V_{\text{dwell}} / h_{\text{floor}}$。

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2.2 Business（每办公室/企业单元 + 单 AP 覆盖面积）

论文在 business 侧用"非住宅楼面面积 → 分配给企业规模 → Wi-Fi 覆盖楼面 → 除以单 AP 覆盖面积"来推 AP 数： Wi-Fi Wardriving Studies Must A...
我们把它压缩成最简单可用版本：

设办公楼面占比 $\eta_{\text{bus}} \in [0,1]$（或用 1−η_res），每个办公室单元平均面积 A_{office}（m²/办公室），每办公室平均 AP 数 κ_{bus} 或者单 AP 覆盖面积 C_{AP}（m²/AP）。二选一即可。

版本B1：按"办公室单元数"计

$$N_{\text{office}} = \sum_i \frac{\eta_{\text{bus}} A_i^{\text{floor}}}{A_{\text{office}}}$$

$$N_{\text{bus}} = \kappa_{\text{bus}}\, p_{\text{bus}}\, N_{\text{office}}$$

版本B2：按"单 AP 覆盖楼面面积"计（更贴合论文 business 公式结构）

$$A^{\text{bus,floor}} = \sum_i \eta_{\text{bus}} A_i^{\text{floor}}$$

$$N_{\text{bus}} = \alpha\, \frac{A^{\text{bus,floor}}}{C_{\text{AP}}}$$

其中 $\alpha \in (0,1]$ 是"并非全部楼面都有 Wi-Fi 覆盖"的比例（论文明确指出这个未知比例参数的存在）。
Wi-Fi Wardriving Studies Must A...

3) 从 AP 数得到 PPP 强度（你背景 PPP 需要的就是这个）

设你的仿真区域面积为 A_{region}（km²）：

$$\lambda_{\text{wifi}} = \frac{N_{\text{wifi}}}{A_{\text{region}}} \quad [\text{AP}/\text{km}^2]$$

如果你要分 2.4 / 5.8 两类：

$$\lambda_{2.4} = f_{2.4}\lambda_{\text{wifi}}, \qquad \lambda_{5.8} = (1 - f_{2.4})\lambda_{\text{wifi}}$$

如果你只要"正在发射的干扰 AP"（考虑 activity/thinning）：

$$\lambda_{\text{tx}} = p_{\text{tx}}\lambda$$

5) 最终给你一个“最简公式包”（你可以直接放进代码/论文）

把上面压缩成一页即可：

建筑→楼面：

$$A^{\text{floor}} = \sum_i (W_i L_i) \cdot \left\lfloor \frac{H_i}{h_{\text{floor}}} \right\rceil$$

住宅 AP：

$$N_{\text{res}} = \kappa_{\text{res}} p_{\text{wifi}} \cdot \frac{\eta_{\text{res}} A^{\text{floor}}}{A_{\text{dwell}}}$$

办公 AP（选其一）：

$$N_{\text{bus}} = \kappa_{\text{bus}} p_{\text{bus}} \cdot \frac{\eta_{\text{bus}} A^{\text{floor}}}{A_{\text{office}}} \quad \text{or} \quad N_{\text{bus}} = \alpha \frac{\eta_{\text{bus}} A^{\text{floor}}}{C_{\text{AP}}}$$

PPP 强度：

$$\lambda_{\text{wifi}} = \frac{N_{\text{res}} + N_{\text{bus}}}{A_{\text{region}}}$$

$$\lambda_{2.4} = f_{2.4} \lambda_{\text{wifi}}, \quad \lambda_{5.8} = (1 - f_{2.4}) \lambda_{\text{wifi}}$$

4) 这套模型为什么能“基于该论文”支撑你的思路？

你要写在论文里的支撑点（不复杂、可审稿）：

1. 该论文把 Wi-Fi AP 来源拆成 **households + businesses** 并分别建模。 Wi-Fi Wardriving Studies Must A...
2. business 部分明确用 **非住宅楼面面积（footprint×floors）** 并通过 **“覆盖面积/未知覆盖比例”** 推 AP 数。 Wi-Fi Wardriving Studies Must A... Wi-Fi Wardriving Studies Must A...
3. household 部分用“宽带采用→Wi-Fi 采用”的概率链来得到“有 Wi-Fi 的户数”，你这里把它压缩成 p_{wifi} 作为合并采用率参数。 Wi-Fi Wardriving Studies Must A...
4. 你做的扩展是：把“户/办公室数量”不从统计表读，而是从你生成的城市建筑几何（尤其是体积/楼面）推出来，仍保持“数量/面积 → 密度”的线性结构，符合该论文“从建筑/统计推 Wi-Fi 分布”的方法学精神。

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4.2 □□□□□□□□□□□□

□□□□□□□□ Wu–Zeng–Zhang □□□ UAV □□□□□□(6)□□□□[7][1]

$$P(V)=P0(1+3V^2/U_{tip}^2)+Pi\cdot(\sqrt{(1+V^4/(4v0^4))}-V^2/(2v0^2))^{1/2}+0.5\cdot d0\cdot\rho\cdot s\cdot A\cdot V^3 \quad (9)$$

```
2823:     def _rotor_propulsion_power_w(self, drone_speed_mps: float) -> float:
2824:         """□□□□□□□□□□ P(V)□W□□□□ bit/J □□□□□"""
2825:         v = max(float(drone_speed_mps), 0.0)
2826:         P0 = float(self.power_model['rotor_P0_w'])
2827:         Pi = float(self.power_model['rotor_Pi_w'])
2828:         U_tip = max(float(self.power_model['rotor_U_tip_mps']), 1e-6)
2829:         v0 = max(float(self.power_model['rotor_v0_mps']), 1e-6)
2830:         d0 = float(self.power_model['rotor_d0'])
2831:         rho = float(self.power_model['air_rho'])
2832:         s = float(self.power_model['rotor_s'])
2833:         A = float(self.power_model['rotor_A_m2'])
2834:
2835:         blade_profile = P0 * (1.0 + 3.0 * (v ** 2) / (U_tip ** 2))
2836:         induced_inner = np.sqrt(1.0 + (v ** 4) / (4.0 * (v0 ** 4))) - (v ** 2) / (2.0 * (v0 ** 2))
2837:         induced = Pi * np.sqrt(max(float(induced_inner), 0.0))
2838:         parasite = 0.5 * d0 * rho * s * A * (v ** 3)
2839:         return float(max(blade_profile + induced + parasite, 1e-6))
2840:
2841:     def calculate_energy_efficiency_metrics(self, scenario: Dict) -> Dict[str, float]:
2842:         """
2843:         □□ EE□bit/J□□EE = (Σ_k R_k) / (Σ_k P_prop(V_k))
2844:         □□□“□□□ k □□□□□□□□□□□□ R_k”□□□□□□□□
2845:         """
2846:         if bool(getattr(self, 'tdma_enabled', False)):
2847:             return self.calculate_tdma_energy_efficiency_metrics(scenario)
2848:         drones = np.asarray(scenario.get('drone_positions', np.zeros((0, 3))), dtype=float)
2849:         stations = np.asarray(scenario.get('station_positions', np.zeros((0, 3))), dtype=float)
2850:         speeds = np.asarray(scenario.get('drone_speeds', np.zeros((len(drones),))), dtype=float)
2851:         if drones.size == 0 or stations.size == 0:
2852:             return {'sum_rate_bps': 0.0, 'sum_propulsion_w': 0.0, 'EE_bpj': 0.0}
2853:
2854:         sum_rate_bps = 0.0
2855:         sum_prop_w = 0.0
2856:         for drone_idx in range(int(len(drones))):
2857:             rate_list = []
2858:             for station_idx in range(int(len(stations))):
2859:                 comp = self.calculate_power_margin_components(
2860:                     drones[drone_idx],
2861:                     stations[station_idx],
2862:                     scenario,
2863:                     drone_idx=int(drone_idx),
2864:                 )
2865:                 rate_list.append(float(comp.get('R_bps', 0.0)))
2866:             mean_rate_bps = float(np.mean(rate_list)) if rate_list else 0.0
2867:             v = float(speeds[drone_idx]) if drone_idx < len(speeds) else 0.0
2868:             sum_rate_bps += mean_rate_bps
2869:             sum_prop_w += self._rotor_propulsion_power_w(v)
2870:
2871:         ee_bpj = float(sum_rate_bps / max(sum_prop_w, 1e-12))
2872:         return {'sum_rate_bps': float(sum_rate_bps), 'sum_propulsion_w': float(sum_prop_w), 'EE_bpj': ee
2873:
2874:     def _build_time_grid(self) -> Tuple[np.ndarray, float]:
2875:         """
2876:         □□□□□□□□t[n]=n·Δt□n=0..N-1□
2877:         □□□□□□□□□□□□□□□□TDMA □□/□□□□
2878:         """
2879:         t_total = float(self.user_params.get('traj_total_time_s', 20.0))
2880:         dt = float(self.user_params.get('traj_dt_s', 1.0))
2881:         t_total = max(t_total, 1e-6)
2882:         dt = max(dt, 1e-6)
2883:         n = int(max(1, round(t_total / dt)))
2884:         t = np.arange(n, dtype=float) * dt
2885:         return t, float(dt)
2886:
2887:     def _trajectory_positions_discrete(self,
2888:                                        drone_idx: int,
2889:                                        scenario: Dict,
2890:                                        t: np.ndarray) -> Tuple[np.ndarray, np.ndarray]:
2891:         """
```

□□□□□□□□ q[n] □□□□□ V[n]□
- hover: q □□□V=0

□□□ □□□□□□□□□□□□□□□□

5.1 □□□□□ CDI

$$\text{Margin} = \text{Pr} - (\text{N} + \text{I} + \text{Threshold}) \quad (10)$$

$$\text{CDI} = 1 - \text{mean_i}(\ \text{clip}(\max(0, -\text{Margin_i})/\text{Deficit_ref}, 0, 1)\ ) \quad (13)$$

```
    def calculate_cdi_components(self,
                                 drone_pos: np.ndarray,
                                 station_pos: np.ndarray,
                                 scenario: Dict,
                                 drone_idx: Optional[int] = None) -> Tuple[float, float, float]:
        """
        □□□□□□□□□□CDI□□□□
        - D_link = max(0, Λ - PL_max)  □□□□□□□□□
        - D_int  = I = Σ h_i l(r_i)   □□□□□□□□□□□□□□
        - CDI    = w_L·D_link + w_I·g(D_int)
        """
        lambda_db = float(self.calculate_atg_mean_path_loss_db(drone_pos, station_pos, scenario.get('bui
        d_link = float(max(0.0, lambda_db - float(self.cdi_pl_max_db)))

        interference_mw: Optional[float] = None
        if drone_idx is not None and 'interference_power_mw_per_drone' in scenario:
            try:
                interference_mw = float(scenario['interference_power_mw_per_drone'][int(drone_idx)])
            except Exception:
                interference_mw = None
        if interference_mw is None:
            interference_mw = float(self.calculate_interference_power_mw(
                drone_pos=np.asarray(drone_pos, dtype=float),
                interference_sources=scenario.get('interference_sources', []),
                buildings=scenario.get('buildings', []),
                drone_idx=drone_idx,
            ))

        g_int_db = self._cdi_intensity_mapping_db(interference_mw)
        cdi = float(self.cdi_w_link * d_link + self.cdi_w_int * g_int_db)
        return d_link, float(interference_mw), cdi

    def calculate_comm_degradation(self,
                                   drone_pos: np.ndarray,
                                   station_pos: np.ndarray,
                                   scenario: Dict,
                                   drone_idx: Optional[int] = None) -> float:
        """
        □□□□□□ ∈[0,1]□
        - legacy□□□□□□□□□□□ + □□/□□□
        - cdi□□□□ CDI □□□□□ 1-exp(-CDI/scale) □□□ [0,1]
        """
        model = str(getattr(self, 'degradation_model', 'margin')).strip().lower()
        if model in ('legacy', 'heuristic', 'old', 'power_ratio'):
            base = float(self.calculate_power_intensity_change(drone_pos, station_pos, scenario, drone_i
            return float(self._blend_u2u_comm_degradation(base, scenario, drone_idx=drone_idx))

        if model in ('margin', 'power_margin', 'm'):
            comp = self.calculate_power_margin_components(drone_pos, station_pos, scenario, drone_idx=dr
            m_db = float(comp['M_db'])
            theta = float(getattr(self, 'margin_threshold_db', 0.0))
            slope = max(float(getattr(self, 'margin_slope_db', 1.0)), 1e-6)
            # comm_deg □□□□□□□□□ M □□□□□□□□□□ 1
            comm_deg = 1.0 / (1.0 + np.exp((m_db - theta) / slope))
            return float(self._blend_u2u_comm_degradation(float(np.clip(comm_deg, 0.0, 1.0)), scenario,

        _, _, cdi = self.calculate_cdi_components(drone_pos, station_pos, scenario, drone_idx=drone_idx)
        scale = max(float(getattr(self, 'cdi_scale_db', 1.0)), 1e-6)
        base = float(np.clip(1.0 - np.exp(-float(cdi) / scale), 0.0, 1.0))
        return float(self._blend_u2u_comm_degradation(base, scenario, drone_idx=drone_idx))

    def _comm_deg_from_margin_db(self, m_db: float) -> float:
        """□□□□□ M[dB] □□□□□□□□□□∈[0,1]□□□□□□□"""
        theta = float(getattr(self, 'margin_threshold_db', 0.0))
        slope = max(float(getattr(self, 'margin_slope_db', 1.0)), 1e-6)
        comm_deg = 1.0 / (1.0 + np.exp((float(m_db) - theta) / slope))
        return float(np.clip(comm_deg, 0.0, 1.0))

    def _u2u_comm_degradation_for_drone(self, scenario: Dict, drone_idx: int) -> Optional[float]:
```

```
        """
        UAV↔UAV □□□□□“□□□□”□□□□□□□□□ I_aer□□
        
        □□□□□□□ U2U □□□□□□calculate_u2u_power_margin_components□□
        □□□ UAV □□□□□ U2U □□□□□□□□□□□mean/max □□□□
        """
        if not getattr(self, 'u2u_enabled', False):
            return None
        
        drones = np.asarray(scenario.get('drone_positions', np.zeros((0, 3))), dtype=float)
        if int(drone_idx) < 0 or int(drone_idx) >= len(drones) or len(drones) < 2:
            return None
```

□□□ □□□□□□□□geatpy□

□□□□ geatpy □□□□□□□□□□ NSGA-II□□“□□□□□/□□ + □□□□□□□”□□□□□□□□□□ CDI□□□□□□□□□□□[1]

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